

fp 32

→ latency } reducing
→ storage }
quantizatⁿ errors ✓

floating point - 32 bits
→ small } 16 bits
into

✓ Weights / Gradient / Activations - 32 bits
32 ✓ large ✓

Quantization

Float 16-bit (FP16)

$$2^{5-1} - 1 = 2^{5-1} - 1 = 15$$

Sign
(1 bit)

0

Exponent
(5 bits)

1 0 0 0 0

$2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$$\text{exp} = 2^4 = 16$$

$$\text{bias} = 2^{5-1} - 1 = 2^{5-1} - 1 = 15$$

$$\text{exp} - \text{bias} = 16 - 15 = 1$$

Significand / Mantissa
(10 bits)

1 0 0 1 0 0 1 0 0 0

$2^{-1} \ 2^{-2} \ 2^{-3} \ 2^{-4} \ 2^{-5} \ 2^{-6} \ 2^{-7} \ 2^{-8} \ 2^{-9} \ 2^{-10}$

$$\text{fraction} = 2^{-1} + 2^{-4} + 2^{-7} = .5703125$$

$$(2^0 + .5703125) = 3.140625$$

$$(-1)^{\text{sign}}$$

$$* 2^{(\text{exponent} - \text{bias})}$$

$$* (1 + \text{fraction})$$

$$(-1)^0 \times (2^1)$$

$$* (1 + (2^{-1} + 2^{-4} + 2^{-7})) = 3.140625$$

$$100 \ 1 \ 00 \ 1000$$

$$-1 \ -2 \ -3 \ -4 \ -5 \ -6 \ -7 \ -8 \ -9 \ -10$$

Bias - fixed by floating point format

$$\text{bias} = 2^{b-1} - 1 = 2^{5-1} - 1 = 2^4 - 1 = 15$$

exponent bits (ft 16)

fp16 - fp32

fp32 precise

Float 32-bit (FP32)

0 1000000000 10010001000000111111011011

$$(-1)^0 \times 2^1 \times 1.5707964 = 3.1415927410125732$$

higher precision

Float 16-bit (FP16)

0 100000 100100010000

$$(-1)^0 \times 2^1 \times 1.5703125 = 3.140625$$

lower precision

original value
3.1415927

*(losing info)
error
reduce the space req
faster inference*

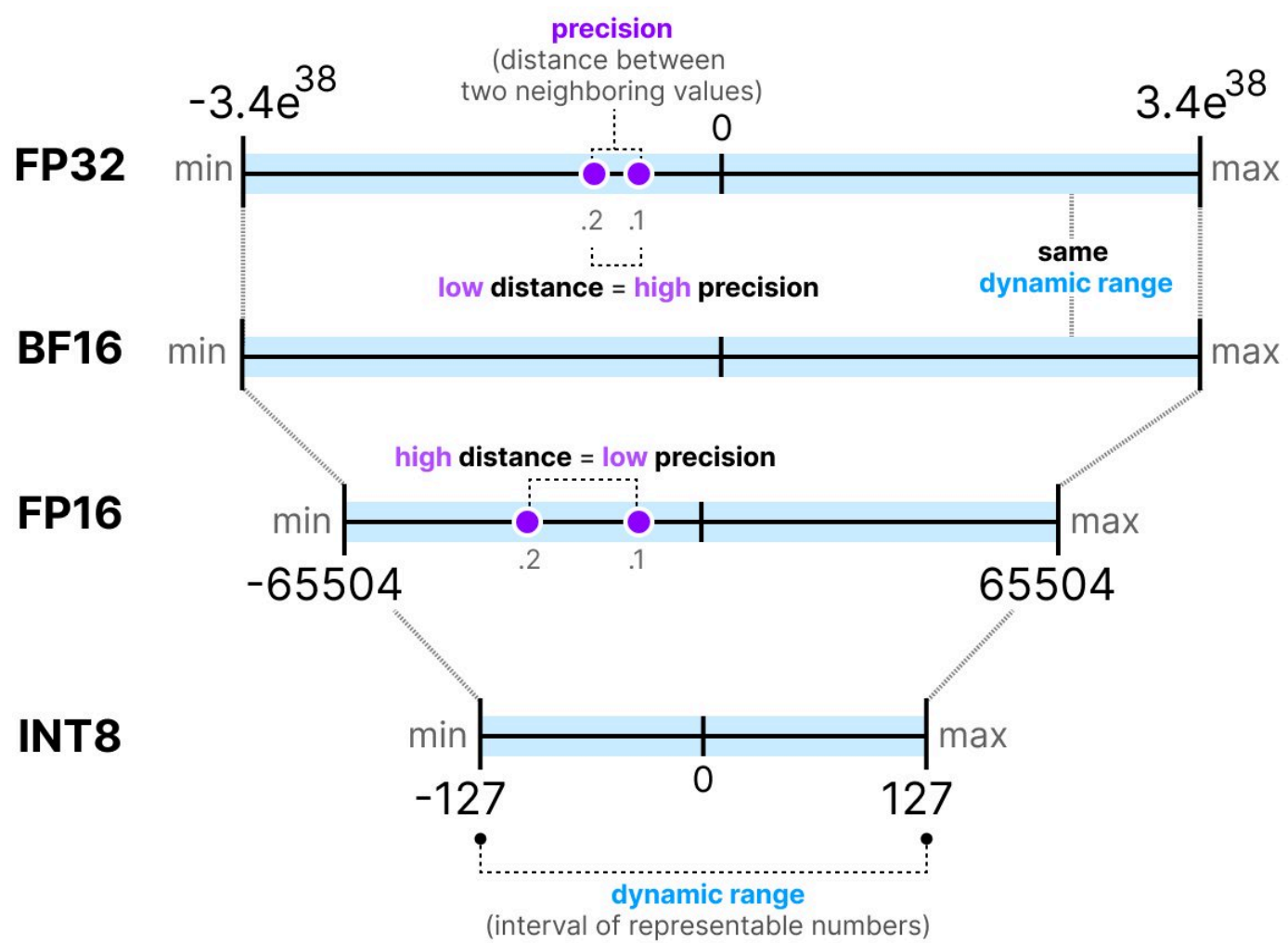
More fraction bits = more detail = higher precision

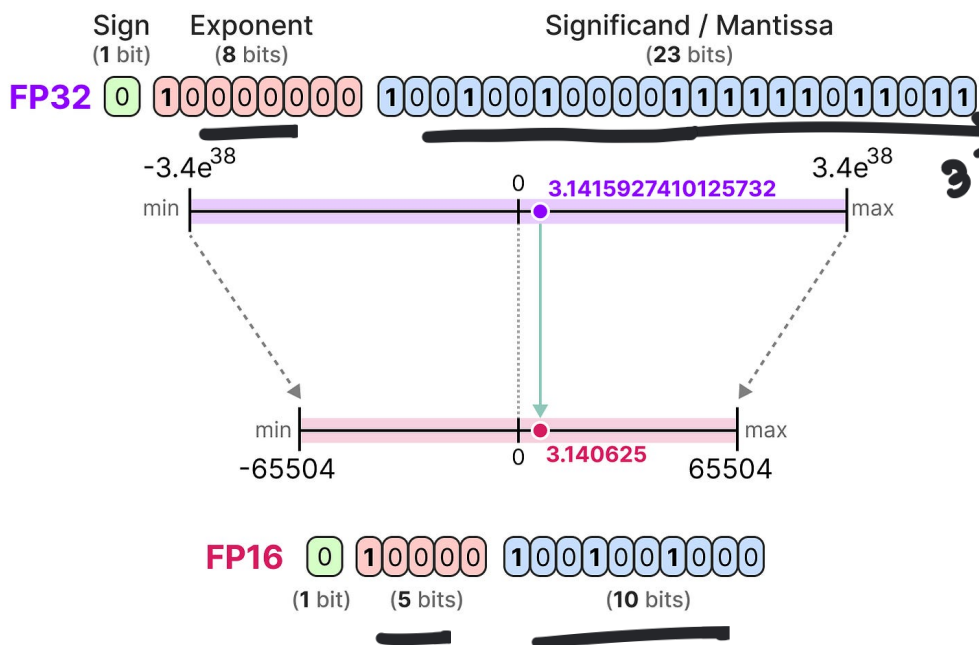
fp32 = keep more decimal places

fp16 = rounding earlier

↑
save memory, faster, some accuracy lost

Brainflood





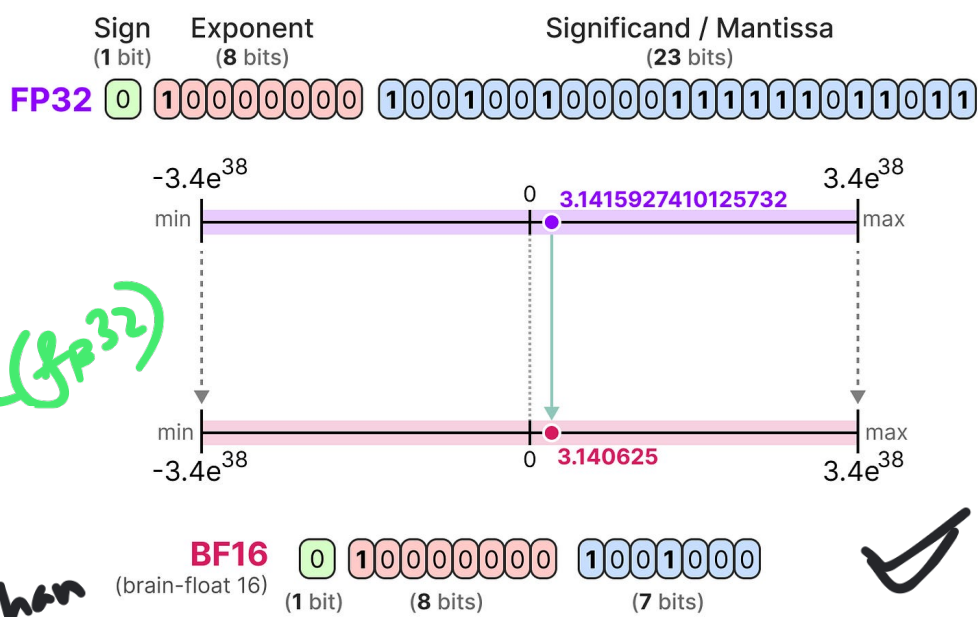
$$3.14 = 3$$

3.14159 2 74101 25732

3.140625

→ Gradient

{ overflow
underflow
collapse

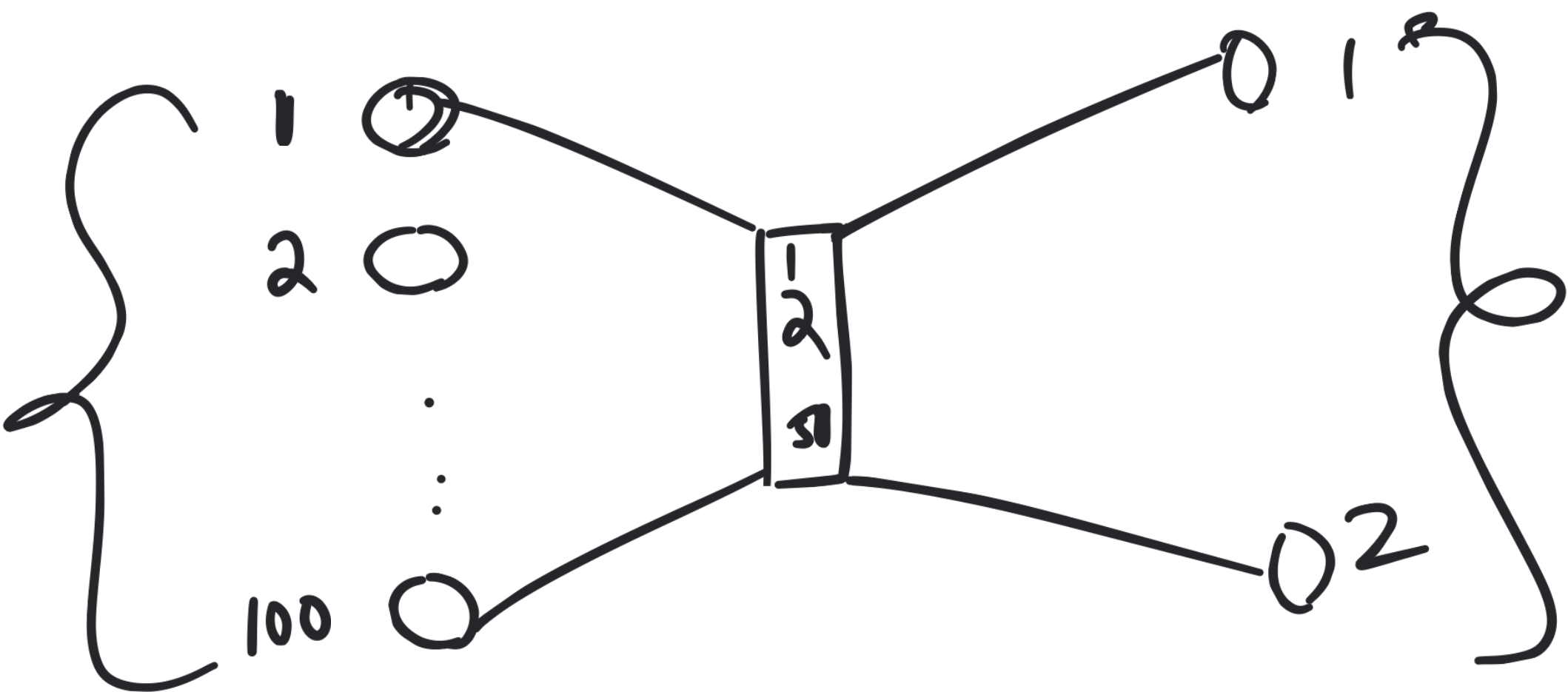


bigger range (fp32)

less precise than fp16

✓ higher range

overflow/
underflow
is less frequent



$\square \square \square \dots \square^k$
 $\{ \square \square \square \square \square^4$
 $\{ \square \square \dots \square^k$

fp16 / bf16

int8

int4

int1

+

bf16

→ lower mem fp32

→ bigger range fp16

→ safe training

→ faster

→ less memory

→ numerical instability ~~loss~~

→ W

→ FP - bfloat16 → less memory
→ faster tensor ops

→ BP - bfloat16 → fast } ✓

→ Wt update grads (bfloat16) → float32
~~bfloat16~~ wh to update float32
float32

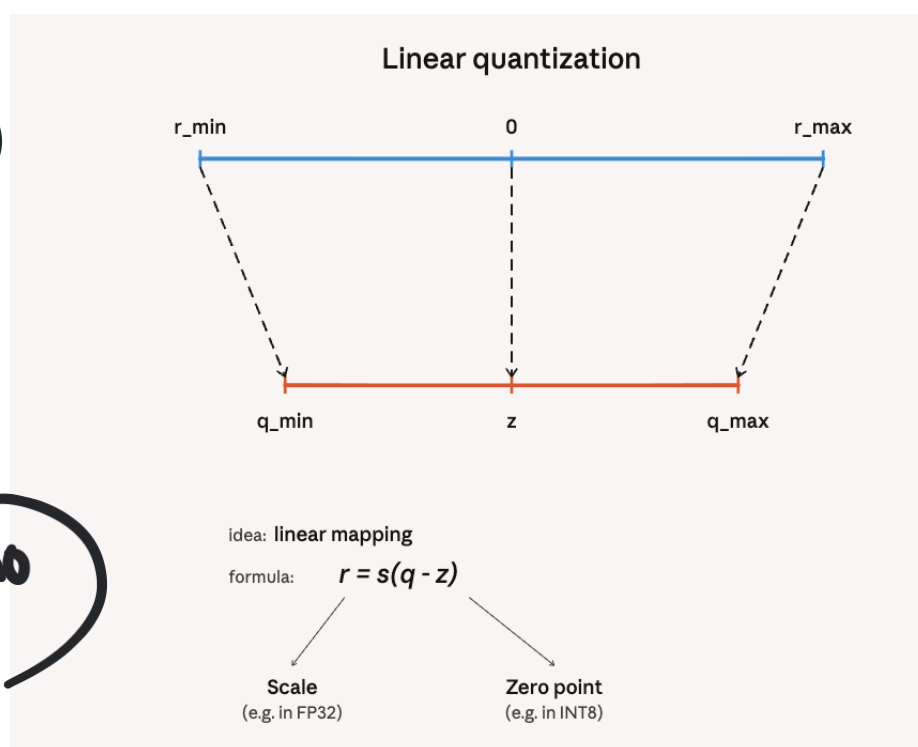
→ next ita - float32 → bfloat16

$$\text{dequantized} = \text{scale}(\text{quantized} - \text{zero point})$$

fp32
↓
 $r = \text{scale}(q - \text{zero point})$
↑
int8

$$\Rightarrow q = \left(\frac{r_{\text{actual}}}{\text{scale}} + \text{zero} \right)$$

↑
int8



← fp32

← fp16/bf16

8 bits

fp 32 $\rightarrow$ int8

Step 1

$$W = \begin{bmatrix} 191.6 & -13.5 & 728.6 \\ 92.14 & 295.5 & -184 \\ 0 & 684.6 & 215.5 \end{bmatrix}$$

Step 2

$$\begin{aligned} r1-min &= \min(W) = -184 \\ r1-max &= \max(W) = 728 \end{aligned}$$

Step 3

$$\text{INT8} \rightarrow [-128, 127]$$

Step 4

$$\text{Scale} = \frac{q_{\text{max}} - q_{\text{min}}}{q_{\text{max}} - q_{\text{min}}}$$

$$= \frac{728 - (-184)}{(127 - (-128))}$$

$$= 912.6 / 255 = 3.58$$

$$\begin{aligned}
 \text{step 5} \quad \underline{\underline{\text{zero point}}} &= q_{\text{-min}} - (n_{\text{-min}}/s) \\
 &= -128 - (-184 / 3.58) \\
 &= -77
 \end{aligned}$$

9:20

$$\text{clip}(\underline{-129}, \underline{128}, 127) = -128$$

$$\text{Quantized} \rightarrow \text{int8} = \text{clip}\left(\text{round}\left(\frac{\text{actual val}}{\text{Scale}} + \text{zero point}\right), -128, 128\right)$$

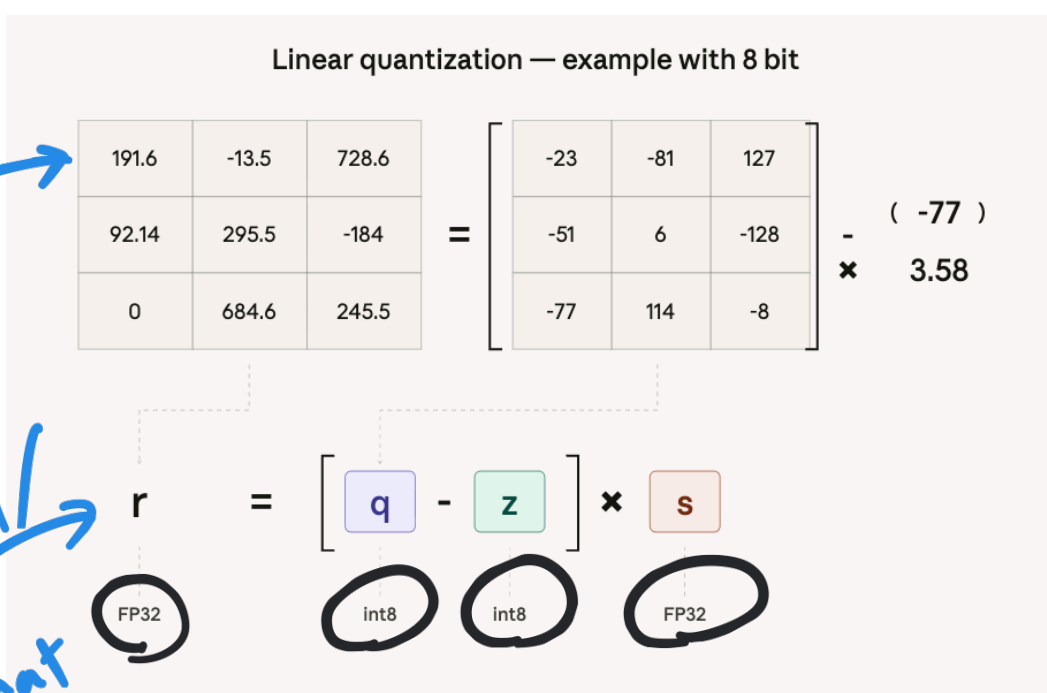
$$\begin{aligned} \text{Quantized} &= \text{clip}\left(\text{round}\left(\frac{191.6}{358} + 77\right), -128, 128\right) \\ &= \text{round} - 23 \end{aligned}$$

$$\text{dequantized} = (-23 - 77) * 9.58 \\ = 193.32$$

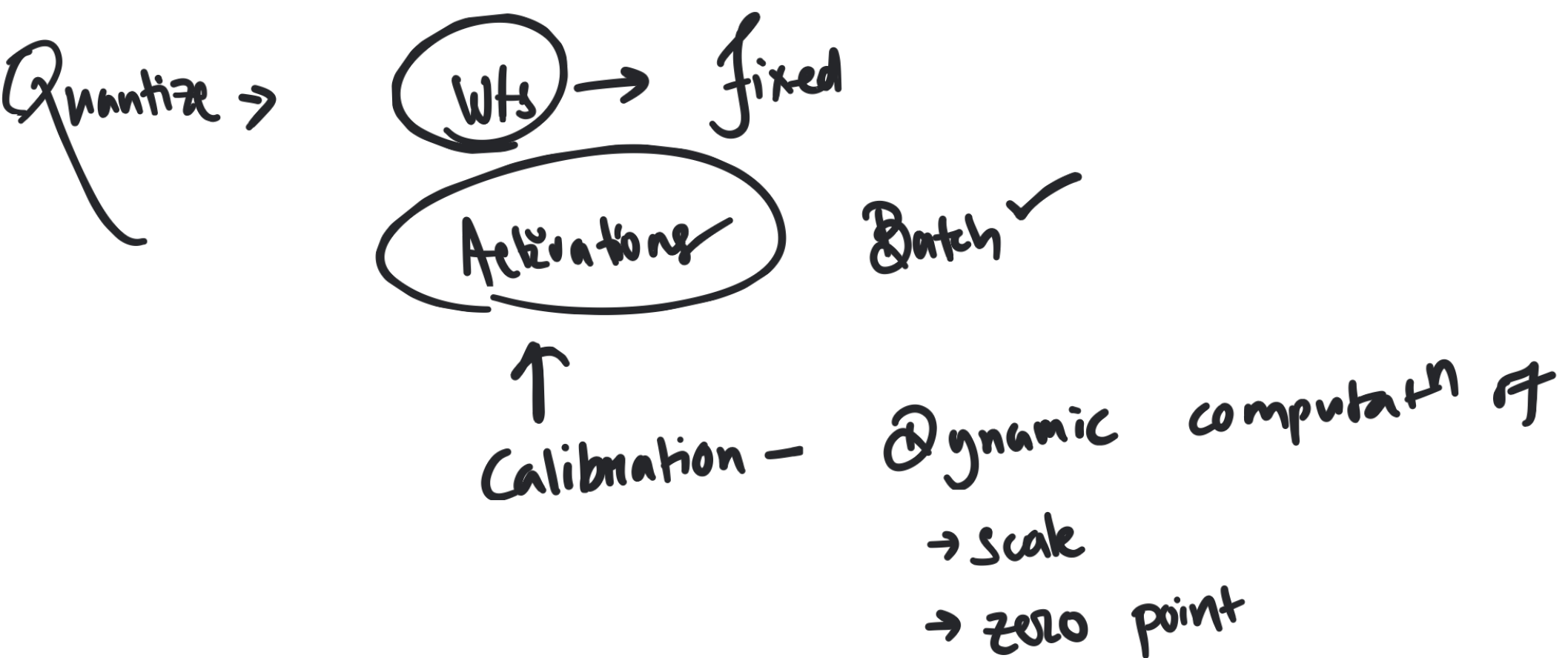
$$193.32 - 191.6 = \text{quantization error}$$

w_{fp32}

dequantized/
original
format

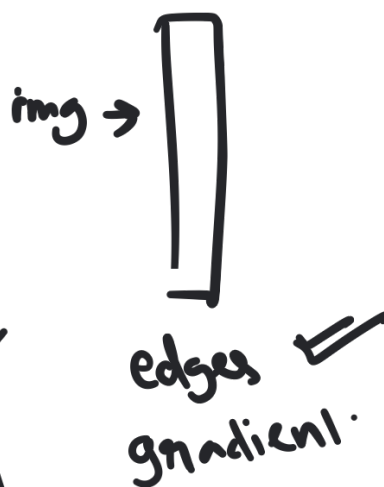


32 bit = 8 bit

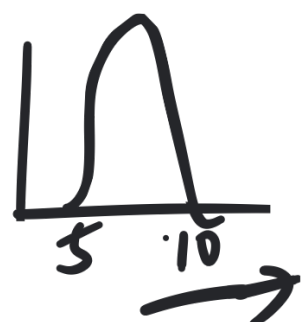


Per channel quantization ✓

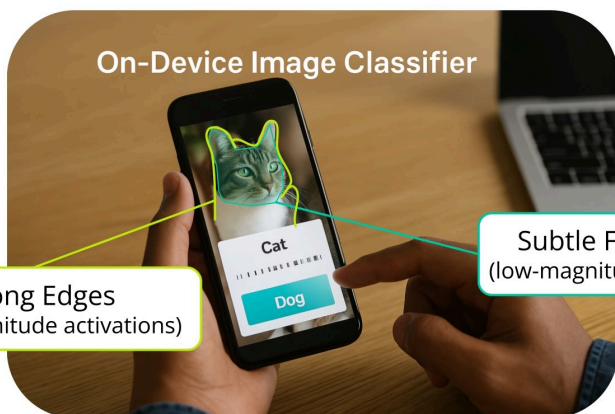
Per tensor quantization



Strong edges



Strong Edges
(high-magnitude activations)



Subtle Fur Texture
(low-magnitude activations)

$28 \times 28 \times 3$

$26 \times 26 \times 32$

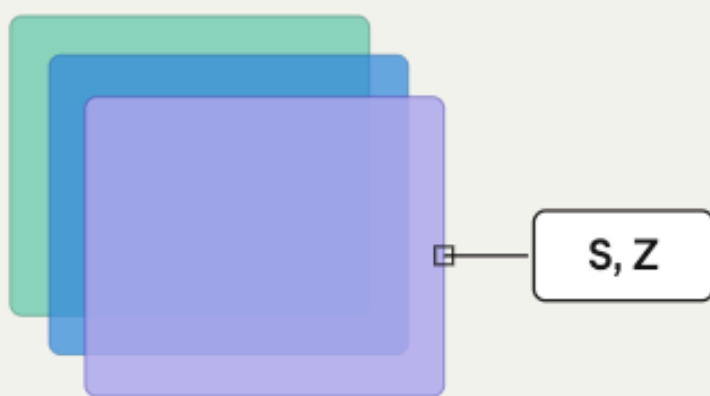
↑
fur texture

$26 \times 26 \times 2$

S
2

Per-tensor vs per-channel quantization

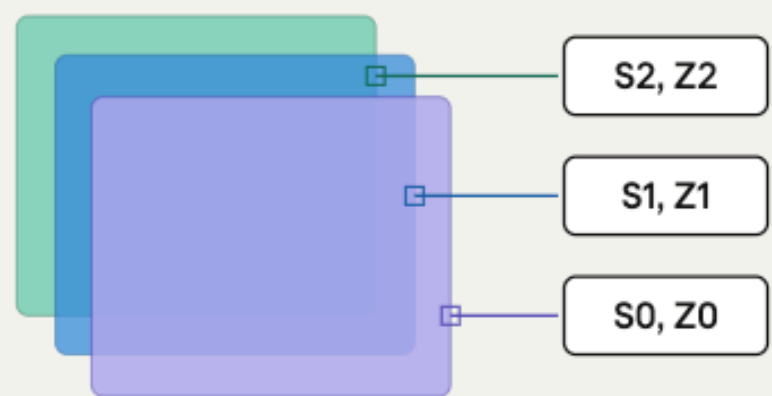
Per-tensor quantization



Every element uses the same mapping:

- Uses minimal metadata
- No extra compute overhead
- Can introduce significant rounding errors

Per-channel quantization



Unique scale and zero-point per channel:

- Captures unique distribution of values per channel
- Reduces quantization noise, preserving accuracy
- Must store and apply one pair of parameters per channel

Quantization

→ Post Training Quantization

→ Quantization Aware Training

Step 1 $\checkmark$
 $W_{fp30} \xrightarrow{q} W_{fp16}$

Step 2
 $W_{fp16} \xrightarrow{\text{dequantize}} W'_{fp32} \checkmark$

$MSE(W_{fp32}, W'_{fp32}) \checkmark$ quantization error



3.145 $\xrightarrow{a}$ 3

3 $\xrightarrow{dv}$ 3.045

32 bits



16 bits (info loss)



32 bits

QAT Quantization Loss Training

$$W = \begin{bmatrix} 0.55567 & -1.0023 \\ 1.57892 & 0.892345 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0.5 & -1.0 \\ 1.4 & 0.87 \end{bmatrix}$$

→ $W_{\text{Quantize}} = \begin{bmatrix} 0.55400 & -1.000 \\ 1.5700 & 0.892 \end{bmatrix}$ FP